

Assignment – Ames Housing Price Prediction Project

Objective:

Build a complete Machine Learning regression workflow using the Ames Housing dataset. You are expected to perform exploratory analysis, preprocessing, feature engineering, pipeline construction, hyperparameter tuning, and comparative evaluation.

Dataset Understanding

- Load and inspect the dataset.
- Identify the target variable (SalePrice) and describe the features.
- Report dataset shape, data types, duplicates, and missing values.

Exploratory Data Analysis (EDA)

- Perform manual EDA using pandas, matplotlib, and seaborn.
- Include descriptive statistics and visualizations.
- Analyze distributions, outliers, correlations, and relationships with SalePrice.
- Perform automated EDA using **ydata-profiling** and export the HTML report.
- Compare insights from manual and automated EDA.

Data Preprocessing

- Handle missing values.
- Handle duplicates if necessary.
- Separate numerical, nominal categorical, and ordinal categorical features.
- Apply One-Hot Encoding for nominal features.
- Apply Ordinal Encoding for ordinal features.
- Scale numerical features when appropriate and justify the scaler.
- Apply feature transformations where needed (e.g., log transform for SalePrice or skewed features).
- Split the data into train/validation/test sets.

Pipeline & ColumnTransformer Requirements

- Use sklearn Pipeline and ColumnTransformer.
- Create at least THREE pipelines:
 1. A preprocessing pipeline for numerical features.
 2. A preprocessing pipeline for One-Hot encoded categorical features.
 3. A preprocessing pipeline for Ordinal encoded categorical features.
- Combine all preprocessing pipelines into a single ColumnTransformer.
- Build complete end-to-end ML pipelines that include preprocessing and the regression model.
- Perform all preprocessing inside the pipeline (avoid data leakage).

Model Development

- Train at **least one other regression** model beside the SVR (e.g., Linear Regression, KNN).
- Use GridSearchCV or RandomizedSearchCV for hyperparameter tuning.

Support Vector Machine (SVM) – Suggested Grid

```
param_grid = {  
    'C': [0.1, 1, 10, 100],  
    'kernel': ['rbf', 'poly'],  
    'gamma': ['scale', 'auto', 0.01, 0.001],  
    'degree': [2, 3]  
}
```

Note:

You may reduce the grid parameters for faster training

Explain how kernel choice, C, gamma, and degree influence model behavior.

- Explain the purpose of the tuned hyperparameters.

Evaluation & Comparison

- Evaluate models using MAE, MSE, RMSE, and R^2 .
- Compare training vs validation performance.
- Discuss overfitting/underfitting.
- Select and justify the final recommended model.
- Interpret the most important features when applicable.

Deliverables

- Colab link of your notebook
- Exported HTML report from ydata-profiling
- Final PDF report (maximum 3 pages)
- requirements.txt

Submission Guidelines

- Organize code clearly.
- Document assumptions and preprocessing decisions.
- Ensure reproducibility.
- Submit before the deadline. **Wednesday [15/7/26]**